

# Bode Plot & Simple Filter Design

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## Abstract:

A first-order low-pass filter with a voltage gain of 10 and a cutoff frequency of 3 kHz was designed and verified in LTspice using an LT1630 operational amplifier in a non-inverting configuration. Simulations included the frequency response, step response, and sinusoidal responses at 30 Hz, 3 kHz, and 300 kHz. The LT1368 op-amp was initially used but replaced due to signal attenuation. The LTspice results closely matched the theoretical transfer function, with minor deviations due to op-amp limitations.

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## Design:

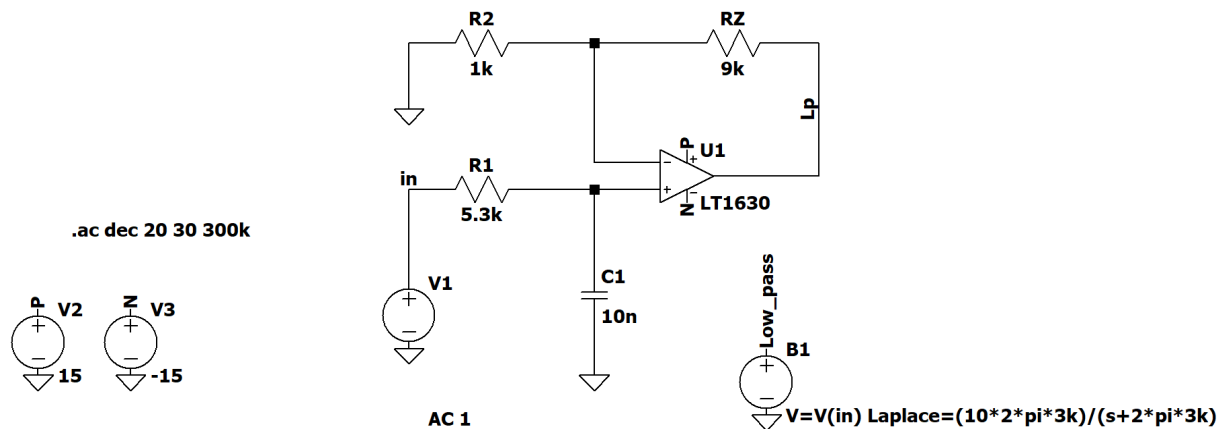


Figure 1: A first-order low-pass filter with a gain of 10 and a cutoff frequency of 3 kHz

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{A_v}{1 + sRC} \quad (\text{eq. 1})$$

Where:

$$A_v = 1 + \frac{R_f}{R_2} \quad (\text{eq. 2})$$

$$H(s) = \frac{V_{out}}{V_{in}} = \text{gain} \times \frac{\omega_0}{\omega_0 + s} \quad (\text{eq. 3})$$

$$\omega_0 = 2\pi f_0$$

$$f_0 = \frac{1}{2\pi \times R_1 C_1}$$

$$\omega_0 = 2\pi \times \frac{1}{2\pi \times R_1 C_1} = \frac{1}{R_1 C_1}$$

Gain = 10

$$H(s) = 10 \times \frac{\frac{1}{R_1 C_1}}{\frac{1}{R_1 C_1} + s} \quad (\text{eq. 4})$$

Cutoff frequency  $f_c$  is defined as:

$$f_c = \frac{1}{2\pi RC} \quad (\text{eq. 5})$$

For the design, the target parameters were  $A_v = 10$  and  $f_c = 3 \text{ kHz}$ . The values were chosen by rearranging equation (5). From this,  $RC = 5.03 \times 10^{-5} \text{ s}$ .

A standard capacitor value  $C_1 = 10 \text{ nF}$

$$R_1 = \frac{1}{2\pi \times f_c \times C_1} = \frac{1}{2\pi(3 \text{ kHz})(10 \text{ nF})} \simeq 5.3 \text{ k}\Omega \quad (\text{eq. 6})$$

A standard  $5.3 \text{ k}\Omega$  resistor was selected.

To achieve the desired gain of 10, the ratio of the feedback resistor from equation (2).

$$A_v = 1 + \frac{R_f}{R_2}$$

Choosing  $R_2 = 10 \text{ k}\Omega$  &&  $R_f = 90 \text{ k}\Omega$

Using the transfer function from equation (4)

$$Y(s) = \frac{1}{s} \times 10 \times \frac{\frac{1}{R_1 C_1}}{\frac{1}{R_1 C_1} + s}$$

Now taking the inverse Laplace

$$y(t) = L^{-1} \left\{ \frac{1}{s} \times 10 \times \frac{\frac{1}{R_1 C_1}}{\frac{1}{R_1 C_1} + s} \right\} = 10(1 - e^{-\frac{1}{R_1 C_1} t})u(t)$$

Where again,  $R_1 \times C_1 = 5.03 \times 10^{-5} s$

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The LT1368 op-amp was tested initially, but the amplitude response exhibited attenuation due to internal compensation capacitance. The LT1630, with a higher gain-bandwidth product (GBW), was substituted and provided accurate performance near the designed cutoff.

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## Verification:

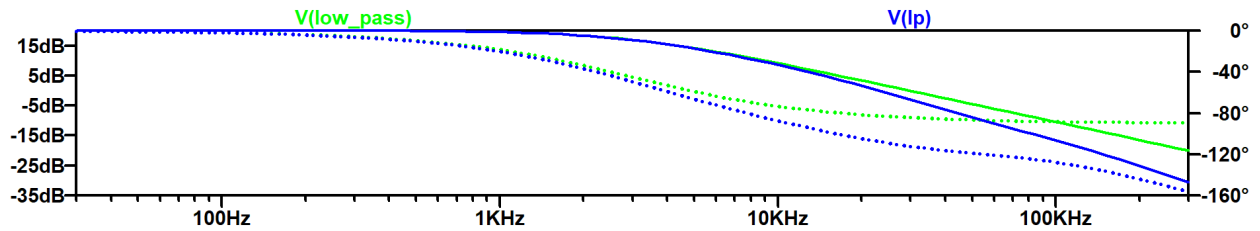


Figure 2. AC Analysis (Bode Plot) using LT1368

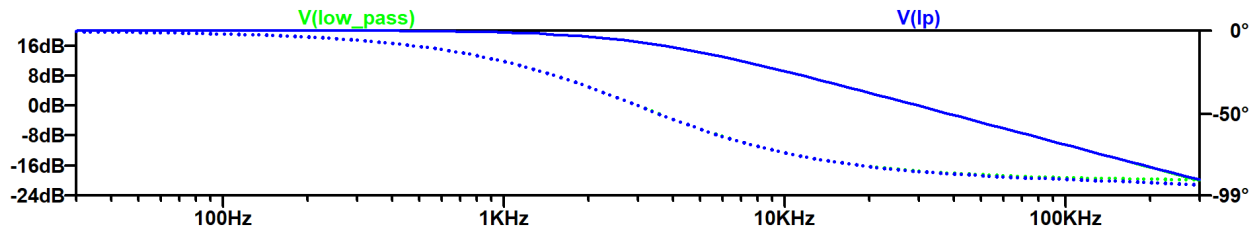


Figure 3. AC Analysis (Bode Plot) using LT1630

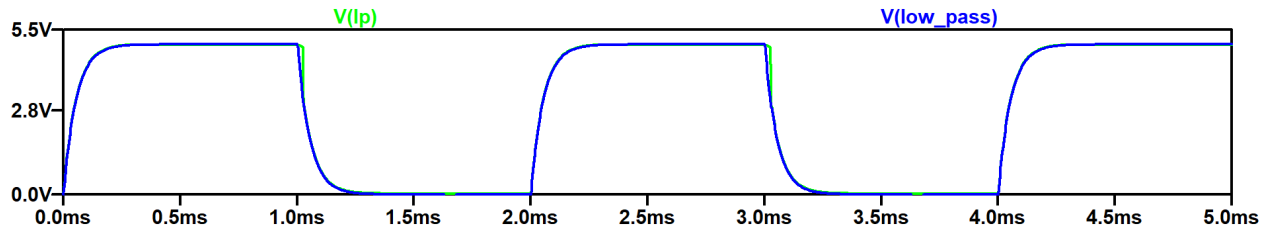


Figure 4. Transient Analysis (Step Response) using LT1630, where  $\rightarrow$  .tran 0 5m 0 10u  
&  $\rightarrow$  PULSE(0 5.5 0 1u 1u 1m 2m)

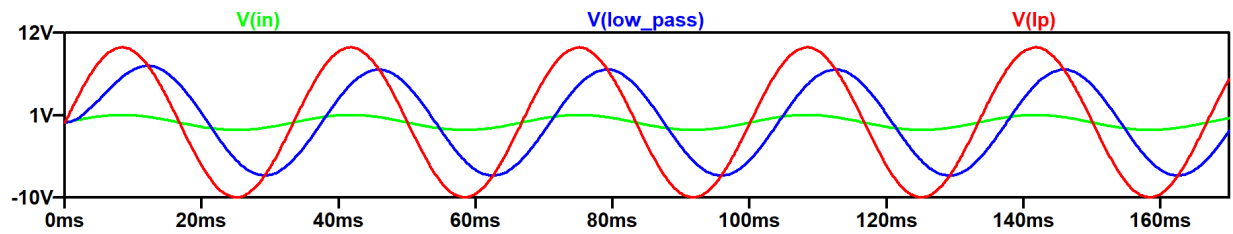


Figure 5. Sine Response using LT1630 of 30 Hz, where  $\rightarrow$  Sine(0 1 30Hz)  
&  $\rightarrow$  .tran 0 170m

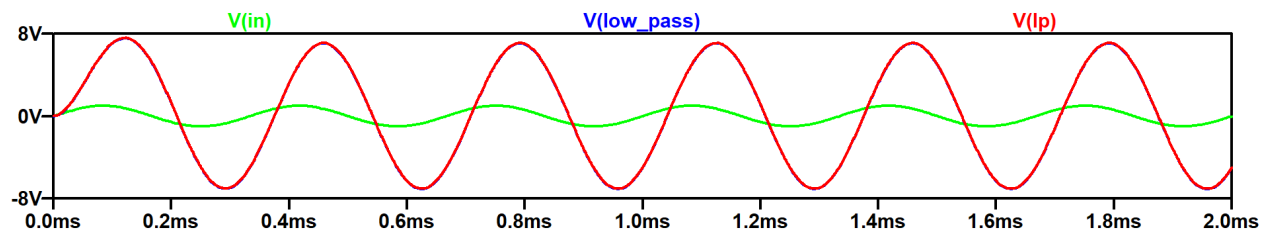


Figure 6. Sine Response using LT1630 of 3 kHz, where  $\rightarrow$  Sine(0 1 3 kHz)  
&  $\rightarrow$  .tran 0 2m

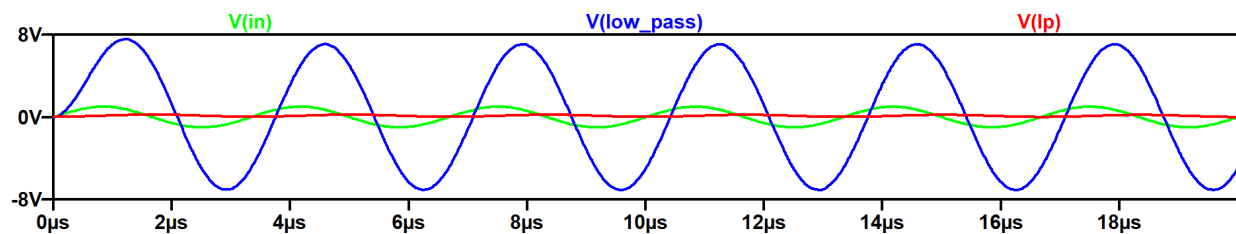


Figure 6. Sine Response using LT1630 of 3 kHz, where  $\rightarrow$  Sine(0 1 300 kHz)  
&  $\rightarrow$  .tran 0 20u

## Sine-Wave Measurements (Transient)

Input amplitude: 1 V peak SINE(0 1 f)

Measured output: **red** trace (Vlp, op-amp output)

Gain in dB:  $20 \times \log_{10}(V_{out\_peak} / V_{in\_peak})$

Phase:  $\varphi = -360 \times (\frac{\Delta t}{T})$ , where  $\Delta t$  is the delay between input and output zero crossings and  $T = \frac{1}{f}$

| Frequency | Period (T)   | Output Peak (V) | Gain (dB) | Phase (Theory)        | Notes              |
|-----------|--------------|-----------------|-----------|-----------------------|--------------------|
| 30 Hz     | 33.3 ms      | 10.0 V          | +20.0 dB  | $\approx -0.6^\circ$  | Correct gain       |
| 3 kHz     | 0.333 ms     | 7.07 V          | +17.0 dB  | $\approx -45^\circ$   | Correct phase lag  |
| 300 kHz   | 3.33 $\mu$ s | 0.3 V           | -10.46 dB | $\approx -89.4^\circ$ | Strong attenuation |

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## Discussion/Summary:

Initial tests with the LT1368 op-amp showed reduced output amplitude near and above the cutoff frequency, consistent with an insufficient gain-bandwidth product (GBW). Replacing it with the LT1630 restored the expected response.

Minor discrepancies between theory and simulation are due to finite op-amp GBW, input offsets, and parasitic capacitances.

At 300 kHz (100 $\times$  the cutoff), theory predicts  $-20$  dB attenuation, while the measured output (0.3 V peak) corresponds to  $-10.46$  dB—acceptable considering non-ideal effects and measurement precision.

The LT1630-based low-pass filter achieved the target 3 kHz cutoff and gain of 10, matching theoretical predictions and demonstrating expected phase behavior of first-order filters.

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## References

[20251010\\_EE10L\\_LOWPASS\\_FILTER\\_AM.asc](#)